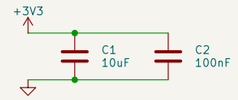
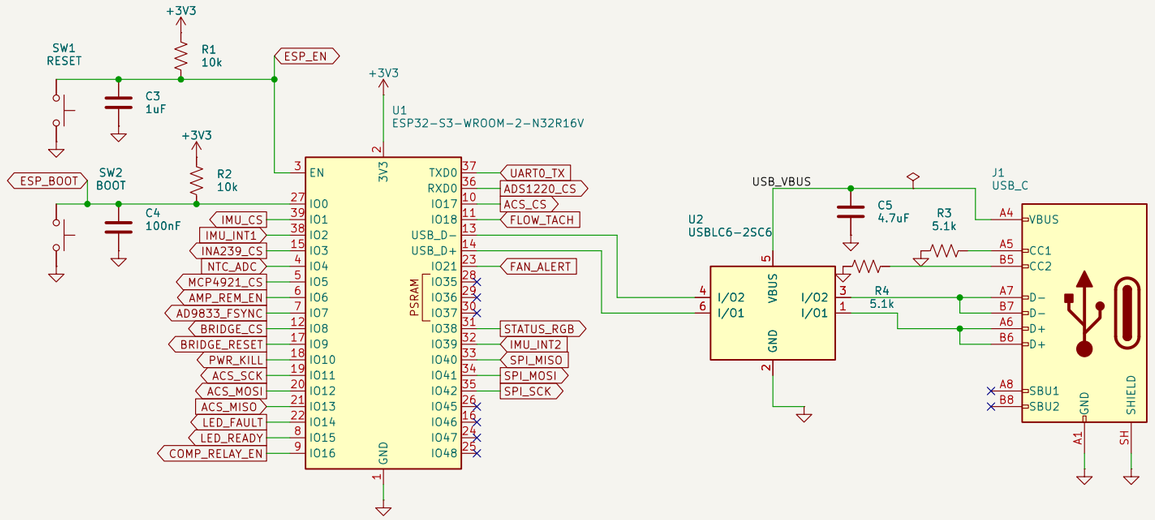
ESP32-S3-WROOM-2-N32R16V: 32 MB octal flash + 16 MB octal PSRAM (GPIO26-37 reserved by module)

GPIO map follows include/config/pin_config.h

ESP32-S3 DECOUPLING CAPACITORS





Power Input File: sheets/power_input.kicad_sch	Signal Generator File: sheets/signal_gen.kicad_sch	Cooling File: sheets/cooling.kicad_sch	Power Button File: sheets/power_button.kicad_sch
Amp Control File: sheets/amp_control.kicad_sch	Cold Head File: sheets/cold_head.kicad_sch	Rails 5V 3V3 File: sheets/rails_logic.kicad_sch	Amp Current File: sheets/amp_current.kicad_sch
IMU File: sheets/imu.kicad_sch	Rails +-15V File: sheets/rails_15v.kicad_sch	System Current File: sheets/system_current.kicad_sch	Indicators File: sheets/indicators.kicad_sch
Compressor Relay File: sheets/compressor_relay.kicad_sch	I2C Bridge File: sheets/i2c_bridge.kicad_sch	Debug File: sheets/debug.kicad_sch	

Cryocooler Controller

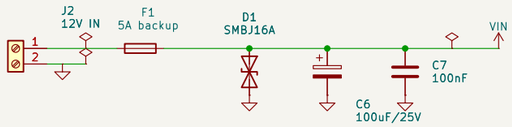
Sheet: /
File: cryocooler.kicad_sch

Title: Cryocooler Controller

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/16

12 V input from a Dell XYK93 (DPS-2000EB class, ~12.2 V, ~160 A available): use the main 12 V rail with PS_ON held low.

F1 is a backup only: add an inline high-breaking-capacity fuse (7.5-10 A) at the PSU end. Reverse polarity: TVS forward-clamps and blows the fuse.



Cryocooler Controller

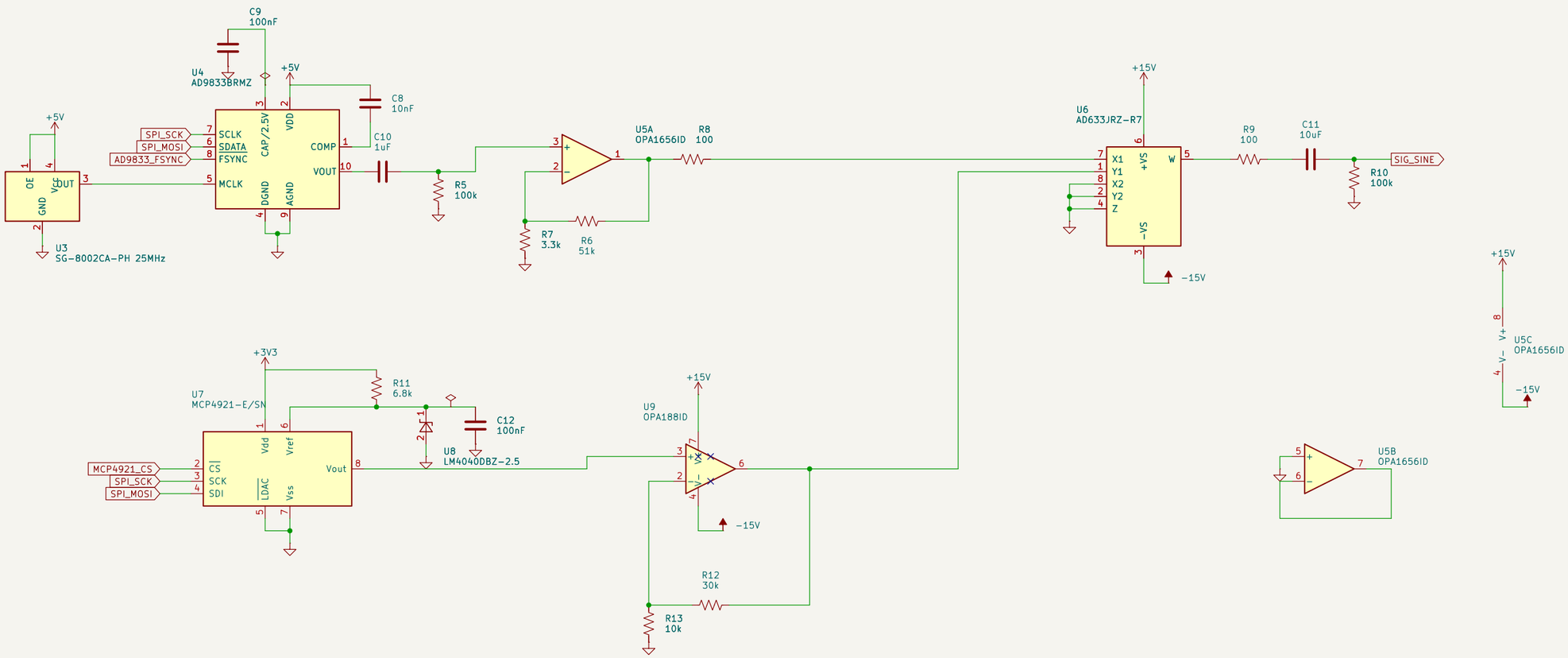
Sheet: /Power Input/
File: power_input.kicad_sch

Title: Power Input

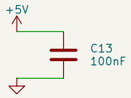
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/16

AD9833 60 Hz sine -> OPA1656 gain stage -> AD633 multiplier; MCP4921 + OPA188 give a 0–10 V amplitude control.

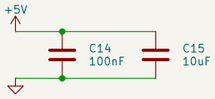
Vctl 0–10 V (2.5 V ref x4) x sine +/−4.9 V pk / 10 = 0..4.9 V pk (9.9 Vpp, 3.5 Vrms) to amplifier input; ZE500.1 RCA input range 0.2–6 V.



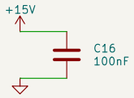
SG-8002CA DECOUPLING CAPACITORS



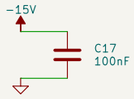
AD9833 DECOUPLING CAPACITORS



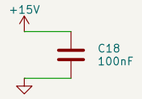
OPA1656 +15V DECOUPLING CAPACITORS



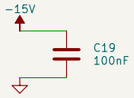
OPA1656 −15V DECOUPLING CAPACITORS



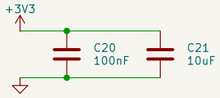
AD633 +15V DECOUPLING CAPACITORS



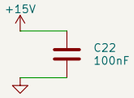
AD633 −15V DECOUPLING CAPACITORS



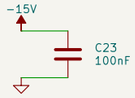
MCP4921 DECOUPLING CAPACITORS



OPA188 +15V DECOUPLING CAPACITORS



OPA188 −15V DECOUPLING CAPACITORS



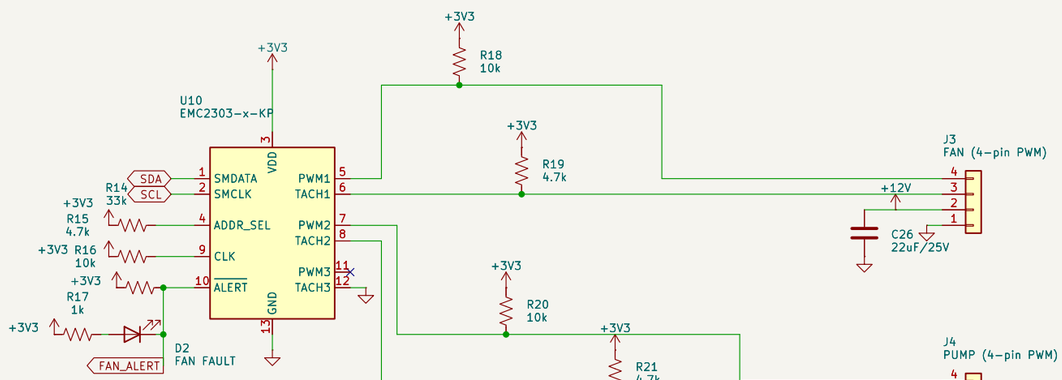
Cryocooler Controller

Sheet: /Signal Generator/
File: signal_gen.kicad_sch

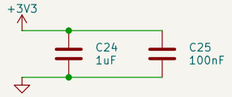
Title: Signal Generator

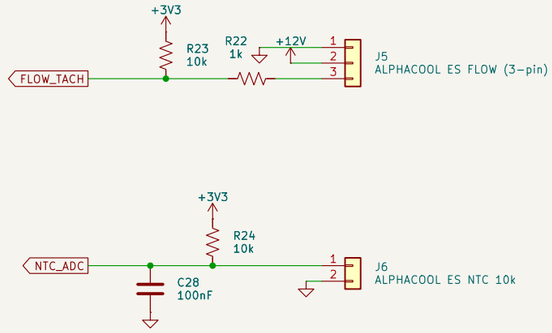
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/16

EMC2303: ADDR_SEL 33k pull-up -> SMBus 0x4D (datasheet Table 5-1); CLK 4.7k pull-up -> default fan drive 0%. ch1 fans, ch2 pump. Alphacool ES: tach -> GPIO18, NTC -> GPIO4.



EMC2303 DECOUPLING CAPACITORS

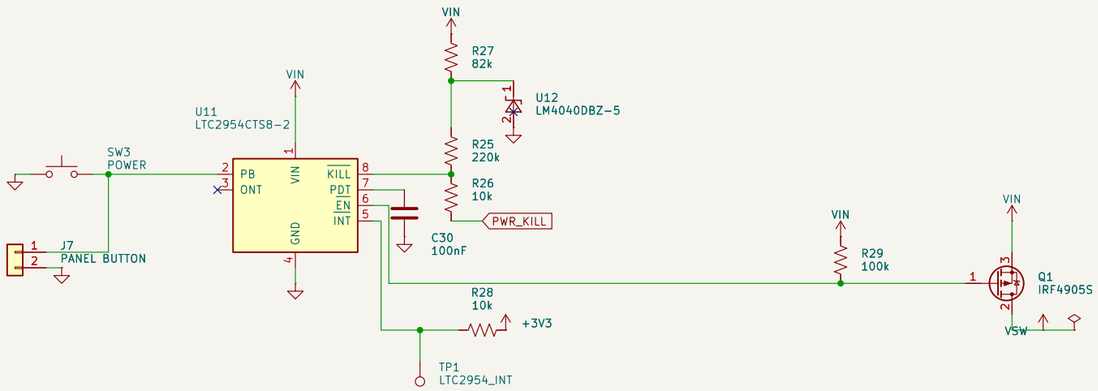




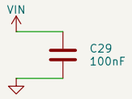
Cryocooler Controller	
Sheet: /Cooling/	
File: cooling.kicad_sch	
Title: Cooling	
Size: A3 | Date:	**Rev: A**
KiCad E.D.A. 10.0.6	Id: 4/16

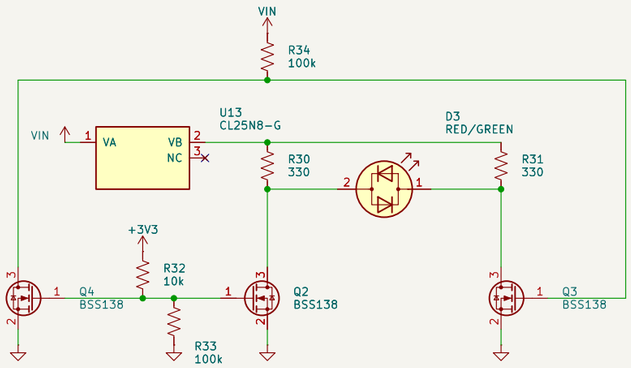
LTC2954-2: tap PB = on, hold = off. EN# drives the P-FET gate; KILL# from GPIO10 (active low, 100 ms pulse).

PWR_ON follows the +3V3 rail: on = green, standby = red. CL25 (25 mA) feeds both legs, ~9 mA reaches the LED, ~25 mA total from VIN in standby



LTC2954 DECOUPLING CAPACITORS

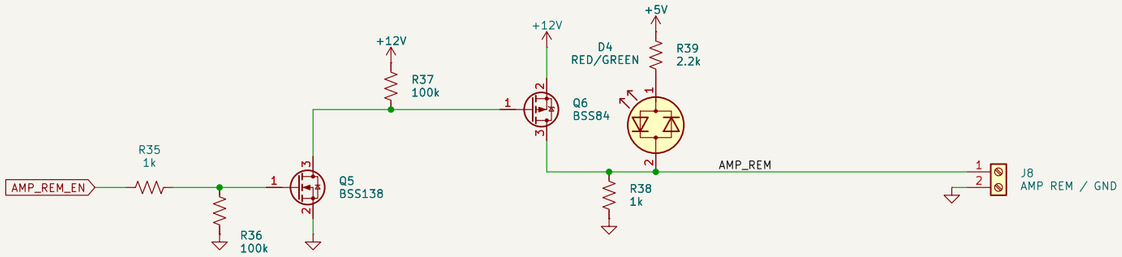




Cryocooler Controller		
Sheet: /Power Button/ File: power_button.kicad_sch		
Title: Power Button		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/16

GPIO6 high -> +12 V on the amplifier REM terminal (P-FET high-side switch). LED: green = on, red = off.

Amplifier: AUDIOZERONE ZE500.1 (external): REM terminal + RCA/line input. Its power leads do not pass through this board.





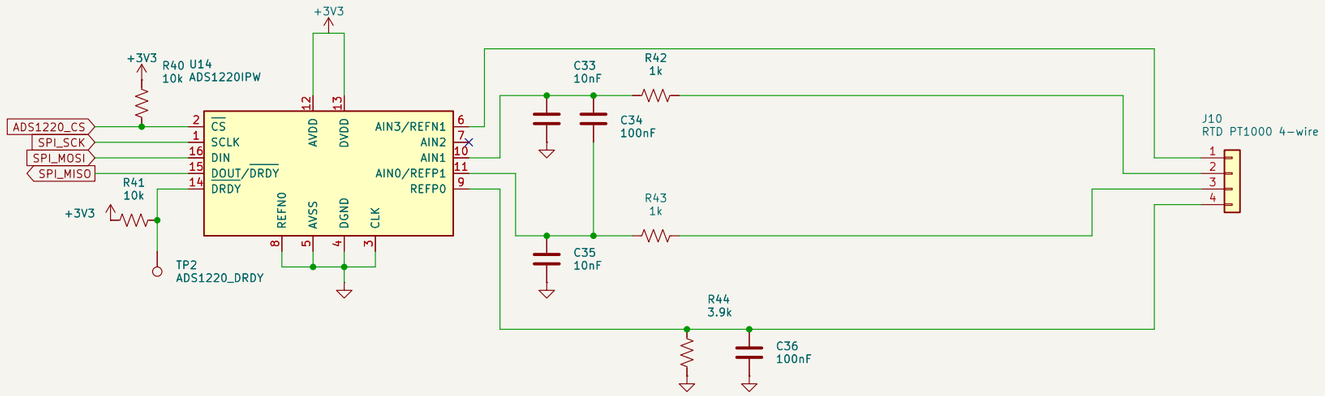
Cryocooler Controller

Sheet: /Amp Control/
File: amp_control.kicad_sch

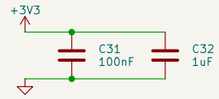
Title: Amp Control

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 6/16

PT1000 4-wire ratiometric: IDAC1 (AIN3) excites, AIN1/AIN0 sense, Rref (3.9k) on REFP0/REFN0. SPI mode 1 on the main bus (CS GPIO44), internal oscillator (CLK tied low). DRDY is not wired to the MCU.



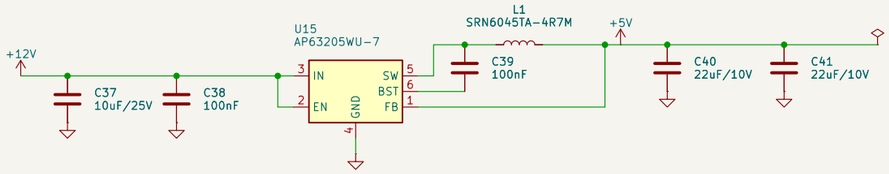
ADS1220 DECOUPLING CAPACITORS

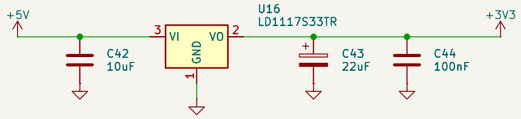


Cryocooler Controller		
Sheet: /Cold Head/ File: cold_head.kicad_sch		
Title: Cold Head		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 7/16

12 V -> 5 V (AP63205 buck, datasheet Fig. 1 / Table 3: 4.7 uH, 10 uF in, 2 x 22 uF out, 100 nF bootstrap) -> 3.3 V (LD1117S33).

AP63205: VIN 3.8–32 V (40 V for 400 ms), EN tied to VIN, FB tied to the fixed 5 V output.





Cryocooler Controller

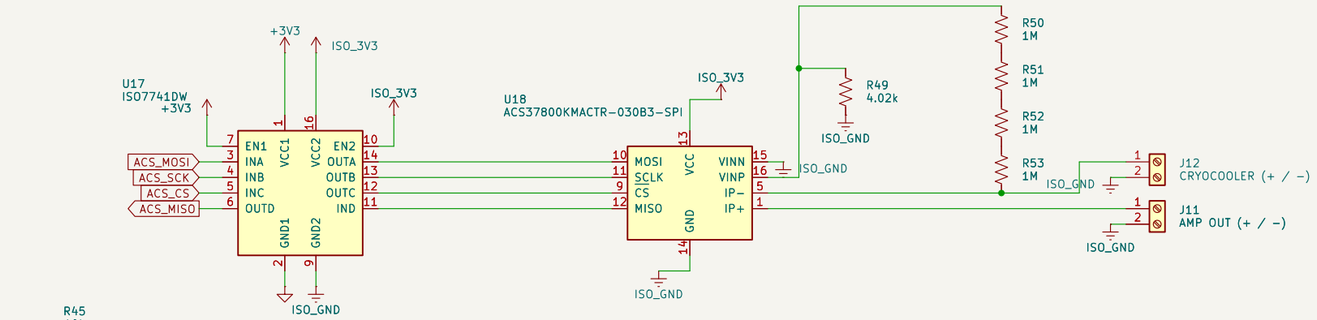
Sheet: /Rails 5V 3V3/
File: rails_logic.kicad_sch

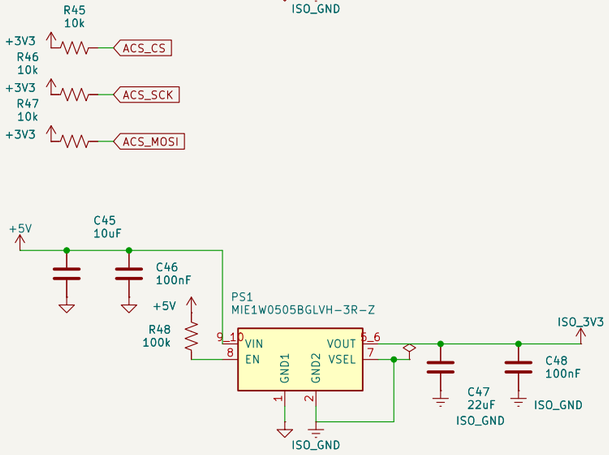
Title: Rails 5V 3V3

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 8/16

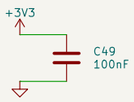
Amplifier output V/I monitor: ACS37800 (SPI) on an isolated 3.3 V domain (MIE1W0505, VSEL=GND2 -> 3.3 V); SPI bus GPIO11/12/13, CS GPIO17.

Mains-class creepage/clearance applies between ISO_GND domain and the logic side on the PCB.

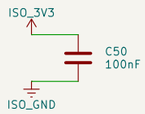




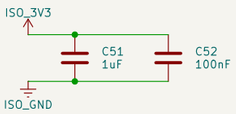
ISO7741 VCC1 DECOUPLING CAPACITORS



ISO7741 VCC2 DECOUPLING CAPACITORS



ACS37800 DECOUPLING CAPACITORS



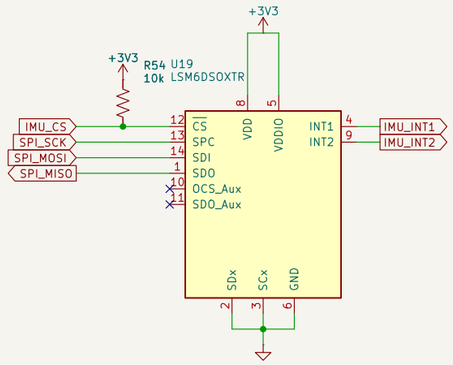
Cryocooler Controller

Sheet: /Amp Current/
File: amp_current.kicad_sch

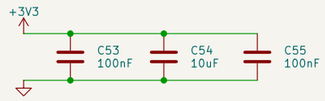
Title: Amp Current

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 9/16

LSM6DSOX in 4-wire SPI mode on the main bus (CS GPIO1, INT1 GPIO2 single-tap, INT2 GPIO39 wake-up).



LSM6DSOX DECOUPLING CAPACITORS



Cryocooler Controller

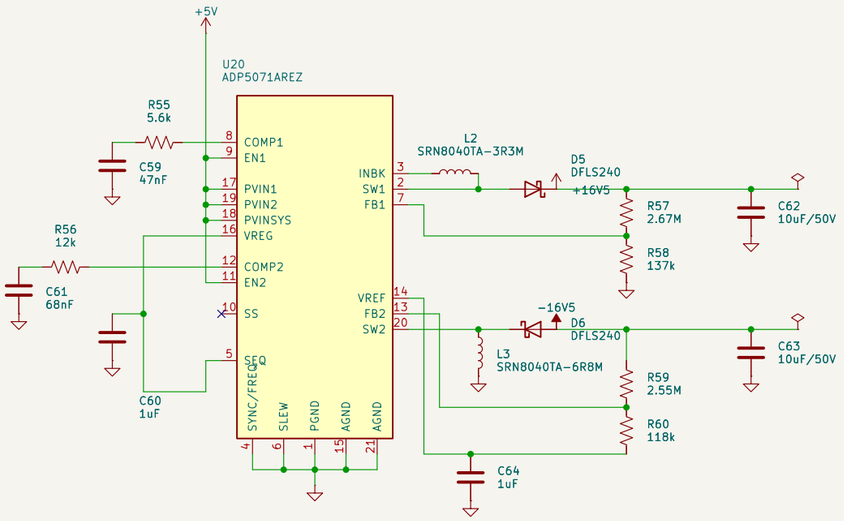
Sheet: /IMU/
File: imu.kicad_sch

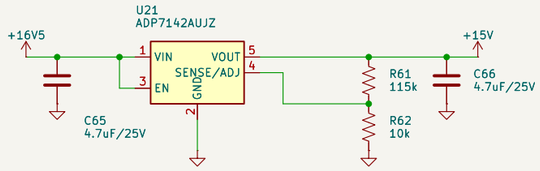
Title: IMU

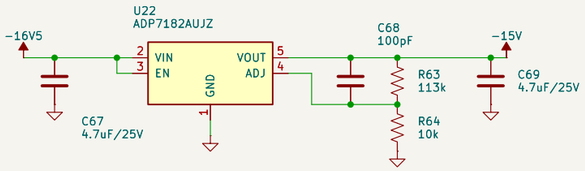
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 10/16

ADP5071AREZ (5 V in): +16.5 V boost and −16.5 V inverting rails; ADP7142 (+15 V) and ADP7182 (−15 V) post-regulate.

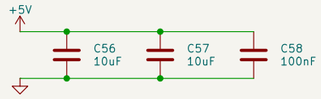
Values from ADP5071 datasheet Fig. 47 / Tables 9–11 (+/−16.5 V output rows rescaled via RFT). ADP7182 EN tied to VIN (auto-start); |VEN| >= 2 V enables.







ADP5071 DECOUPLING CAPACITORS



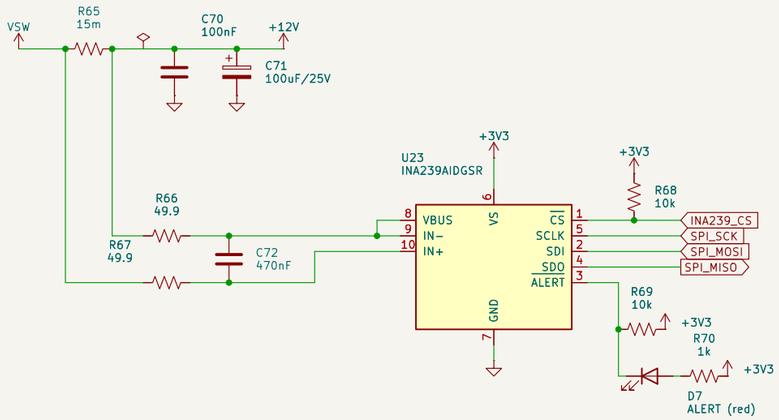
Cryocooler Controller

Sheet: /Rails +−15V/
File: rails_15v.kicad_sch

Title: Rails +−15V

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 11/16

INA239 monitors the switched 12 V rail: 15 mOhm shunt, SPI mode 1 on the main bus (CS GPIO3), +/−163.84 mV range (~10.9 A). ALERT only drives its LED.



3V3 POWER LED

INA239 DECOUPLING CAPACITORS

Cryocooler Controller

Sheet: /System Current/
File: system_current.kicad_sch

Title: System Current

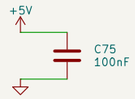
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 12/16

READY (GPIO15) and FAULT (GPIO14) LEDs, active high; WS2812B status LED on GPIO38 via a 74AHCT1G125 3.3 V -> 5 V level shifter (WS2812B VIH = 0.7 x VDD).

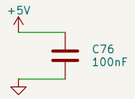


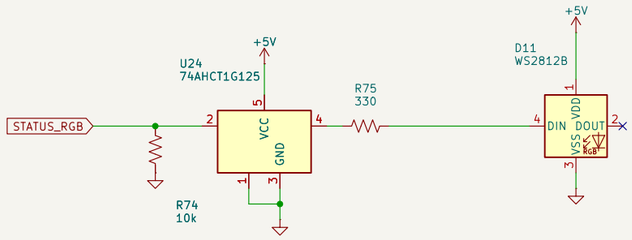


74AHCT1G125 DECOUPLING CAPACITORS



WS2812B DECOUPLING CAPACITORS





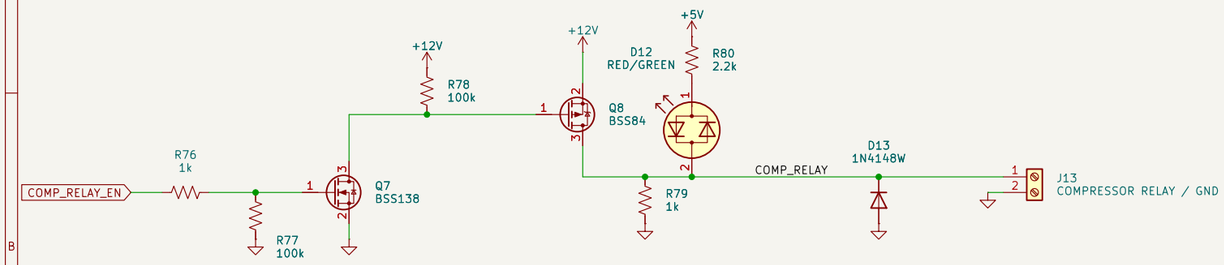
Cryocooler Controller

Sheet: /Indicators/
File: indicators.kicad_sch

Title: Indicators

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 13/16

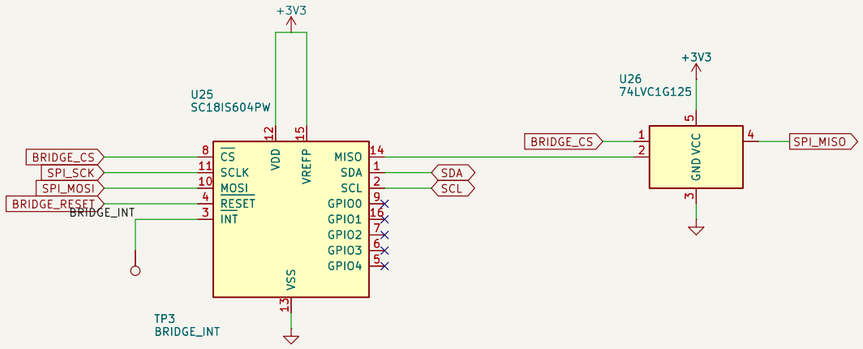
COMP_RELAY_EN (GPIO16) high -> +12 V on the compressor relay terminal (BSS84 high-side switch, 130 mA max: drive an external relay or SSR coil). Flyback diode across the output. LED: green = on, red = off.



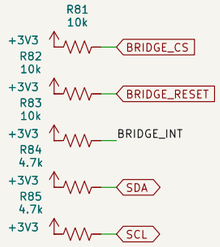
Cryocooler Controller		
Sheet: /Compressor Relay/ File: compressor_relay.kicad_sch		
Title: Compressor Relay		
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 14/16

SC18IS604: SPI mode 3 slave on the main bus (CS GPIO8, RESET GPIO9), I2C controller at ~100 kHz for the EMC2303 (0x4D). INT# is polled through the I2CStat register and only has a pull-up and a test point.

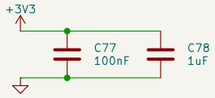
The datasheet does not say MISO is high-impedance with CS high, so a 74LVC1G125 (OE# = CS) gates it onto SPI_MISO.



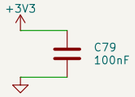
PULL-UPS



SC18IS604 DECOUPLING CAPACITORS



74LVC1G125 DECOUPLING CAPACITORS



Cryocooler Controller

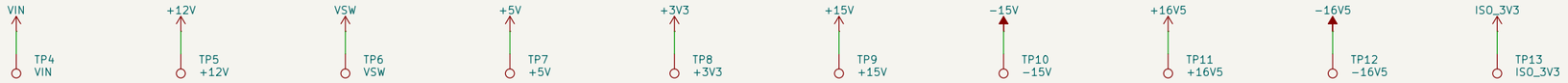
Sheet: /I2C Bridge/
File: i2c_bridge.kicad_sch

Title: I2C Bridge

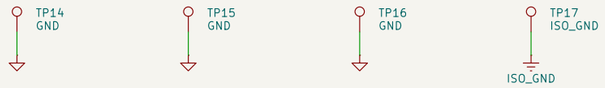
Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 15/16

Debug aids: test points (pad per net, loops for ground) and rail LEDs. Each test point sits on a global label or power symbol; the PCB generator places it beside a pad on the same net.

POWER RAILS



GROUND (scope loops)



SPI BUSES AND CHIP SELECTS



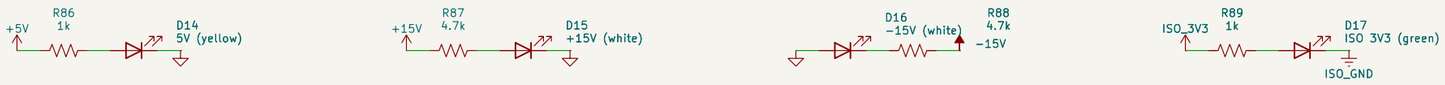
I2C, INTERRUPTS AND CONTROL



ANALOG AND SENSORS



RAIL LEDS



Cryocooler Controller

Sheet: /Debug/
File: debug.kicad_sch

Title: Debug

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 16/16